



Bioenergetic basis of skeletal muscle fatigue

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Energetic demand from high-intensity exercise can easily exceed ATP synthesis rates of mitochondria leading to a reliance on anaerobic metabolism. The reliance on anaerobic metabolism results in the accumulation of intracellular metabolites, namely inorganic phosphate (P_i) and hydrogen (H^+), that are closely associated with exercise-induced reductions in power. Cellular and molecular studies have revealed several steps where these metabolites impair contractile function demonstrating a causal role in fatigue. Elevated P_i or H^+ directly inhibits force and power of the cross-bridge and decreases myofibrillar Ca^{2+} sensitivity, whereas P_i also inhibits Ca^{2+} release from the sarcoplasmic reticulum (SR). When both metabolites are elevated, they act synergistically to cause marked reductions in power, indicating that fatigue during high-intensity exercise has a bioenergetic basis.

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Introduction

Skeletal muscle produces movement by converting chemical energy into mechanical energy through the hydrolysis of ATP. The total energetic demand of this process is the sum of the ATP hydrolyzed for ion transport (SR- Ca^{2+} and Na^+/K^+ ATPases) and the chemo-mechanical transduction of the myosin–actin interaction (myofibrillar ATPase). When the energetic demand of a motor task is low enough that ATP synthesis can be met primarily by oxidative phosphorylation, the mechanical power and force outputs necessary to perform the exercise can be sustained for hours. In contrast, as soon as the exercise intensity exceeds the ATP synthesis rates of oxidative phosphorylation requiring an increased reliance

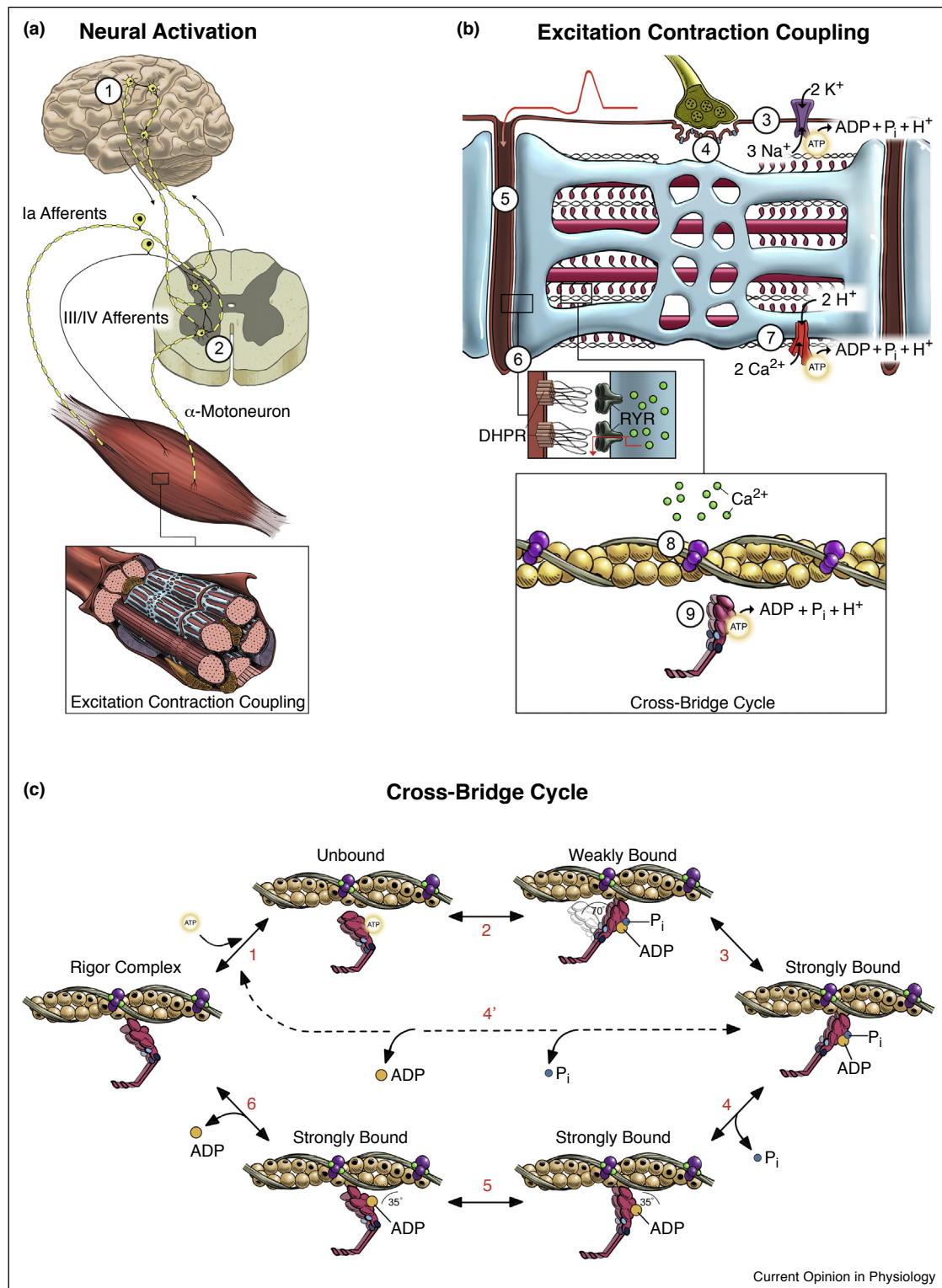
on anaerobic metabolism, force and power become impaired, that is, fatigue develops, and failure to sustain the exercise occurs within seconds to minutes of the onset of contractile activity [1,2]. Any alterations to the mechanics of the contraction, such as contractile frequency, shortening velocity, or duty cycle, will inevitably change 1) the energetic demand and extent the exercise relies on anaerobic metabolism and 2) the rate at which fatigue develops [3]. The observation that the rate of fatigue development has a bioenergetic basis determined by the extent the exercise relies on anaerobic metabolism has been consistently documented in isometric and dynamic contractions [3–5], whole-body and isolated-limb exercises [1,3,6], and in men and women [2].

Although it is generally accepted that fatigue during high-intensity exercise has a bioenergetic basis, identifying the mechanisms by which a reliance on anaerobic metabolism causes fatigue has proven considerably more challenging. This is in large part because fatigue can originate at multiple locations along the motor pathway (Figure 1), and the central nervous system is tightly tuned to the metabolic state of the muscle through sensory feedback from group III/IV afferents [7–9]. A preponderance of evidence, however, indicates that most of the fatigue during high-intensity exercise is due to impaired contractile function within the muscle [4,6], and that this observation is true for healthy young and old men and women [10,11]. Identifying how a reliance on anaerobic metabolism impairs muscle force and power in such a predictable manner is particularly important because in whole-body exercises, such as cycling, ~60–70% of the power outputs generated by the neuromuscular system are non-sustainable and elicit failure within seconds to minutes [1,2]. In this brief review, we integrate findings from *in vivo* to isolated cellular and molecular studies to identify how a reliance on anaerobic metabolism causes fatigue by disrupting contractile function in the muscle.

Anaerobic metabolism impairs contractile function through the accumulation of intracellular metabolites

Intracellular concentrations of ATP ([ATP]) in quiescent skeletal muscle are low (~5–6 mmol/kg wet weight or ~8.2 mM) and could be depleted in less than two seconds during high-intensity exercise. The depletion of ATP would cause the contractile proteins to enter a state of rigor which does not occur *in vivo*; rather, intracellular [ATP] is maintained relatively stable via the synchronized activation of the creatine kinase and adenylate kinase reactions, glycolysis and oxidative phosphorylation. Under the most severe fatigue conditions,

Figure 1



Schematic of potential sites of fatigue along the motor pathway.

Potential sites of fatigue are labeled 1–9 starting with the ability of the nervous system to activate the muscle (a) and progressing to excitation contraction coupling (b) and the cross-bridge cycle (c). During volitional contractions, skeletal muscle is activated via signals originating in the motor cortex (site 1) that are transmitted to the α -motoneurons in the spinal cord (site 2). The output of the polydendritic α -motoneuron is determined by the ensemble synaptic input from thousands of sensory and descending neural pathways, with the group III/IV afferents

intracellular [ATP] rarely falls below 60–70% of resting values in the whole-muscle [12–14], with the most severely depleted muscle fibers reaching ~20% of resting [ATP], or ~1.6–1.8 mM [15]. Even in these severe conditions, the [ATP] has not reached the level necessary to observe impairments in contractile function [12,13]. Thus, barring the unlikely event of a pronounced intracellular compartmentalization of ATP, there is little to no evidence that high-intensity exercise is limited by the rate at which ATP can be synthesized and supplied to the myofibrillar and ion transport ATPases. However, buffering the fall in [ATP] with glycolysis and the creatine kinase and adenylate kinase reactions results in marked disruptions in intracellular homeostasis through the accumulation of ATP hydrolysis metabolites, ADP, inorganic phosphate (P_i), and hydrogen (H^+), the latter causing pH to decrease ($pH = -\log_{10}[H^+]$).

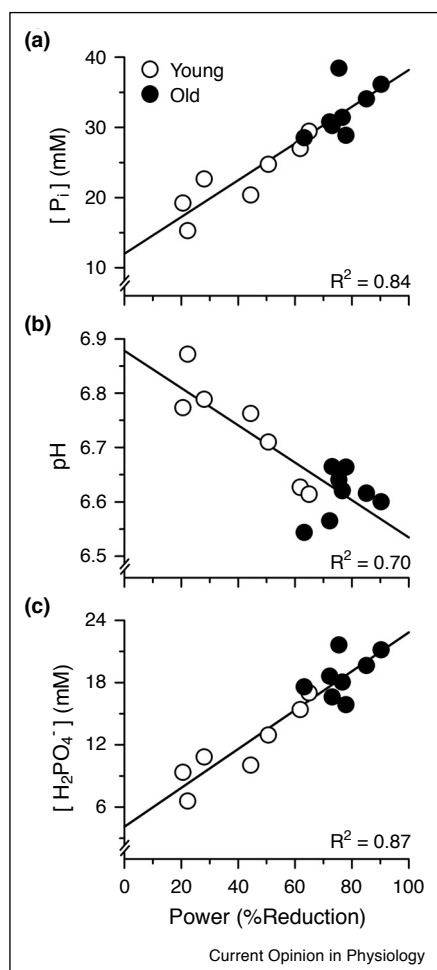
Direct measures of intramuscular bioenergetics via phosphorus nuclear magnetic resonance spectroscopy (^{31}P -MRS) have provided unprecedented detail of the time course and extent of intracellular metabolite accumulation that occurs during high-intensity exercise *in vivo*. In quiescent human skeletal muscle, intramuscular pH is ~7.0, $[P_i]$ ~3–5 mM, and [ADP] only a few μM [16]. At the onset of high-intensity exercise, ATP is synthesized primarily via the creatine kinase reaction resulting in an exponential decline in the concentration of phosphocreatine ([PCr]) with concomitant increases in $[P_i]$ that can reach >30 mM in human skeletal muscle during volitional exercise [17]. Similarly, other than the first few seconds of exercise where the intracellular pH becomes more alkaline by ~0.1 pH units due to PCr hydrolysis ($PCr + ADP + H^+ \leftrightarrow ATP + Cr$) [18], pH declines precipitously and can reach levels between 6.5 to 6.2 during high-intensity exercise [17,19,20], with more severe acidic states possible within regions of the muscle. In contrast, due to the creatine kinase and adenylate kinase reactions, [ADP] remains relatively stable until [PCr] is nearly depleted, after which [ADP] increases by a

few hundred μM , which are levels that do not appear to significantly disrupt contractile function [12]. It is important to note that other compounds accumulate during high-intensity exercise, such as, AMP, IMP, creatine, lactate, extracellular K^+ , Mg^{2+} , and reactive oxygen and nitrogen species, that have been implicated in fatigue; however, with the possible exception of K^+ at high-firing frequencies [21], either 1) do not impair contractile function or do not reach the concentrations during volitional exercise where impaired contractile function occurs, 2) are not correlated with the reductions in force or power, or 3) cause structural damage to proteins that requires days to recover from and arguably can no longer be considered a mechanism of fatigue [13,21,22–24]. Below we present evidence to suggest that a majority of the fatigue during high-intensity volitional exercise can be explained by the multifaceted and synergistic effects of elevated H^+ and P_i .

Although *in vivo* studies are unable to determine whether the accumulation of H^+ and/or P_i causes fatigue, two primary observations suggest that these metabolites play a major role in limiting force and power production during high-intensity exercise. The first is the further the exercise intensity exceeds the level that can be supported by oxidative phosphorylation, the more rapidly H^+ and P_i accumulate in the muscle [5,25] and the decrements in contractile function occur [4]. Accordingly, reducing O_2 availability through inspired hypoxic gas [26] or blood flow occlusion [27] results in a more rapid accumulation of intracellular H^+ and P_i and more rapid impairments in contractile function [28,29]. The second, and perhaps more important observation, is that the extent of intracellular H^+ and P_i accumulation is closely associated with the decrements in 1) force production during volitional [20,30] and electrically evoked isometric contractions [17] and 2) power production during volitional dynamic exercise (Figure 2) [31]. Although these observations do not indicate a causal role for H^+ and P_i in fatigue, they do provide the premise for mechanistic studies employing isolated

(Figure 1 Legend Continued) transmitting signals specific to the metabolic state of the muscle. The propagation of the action potential across the neuromuscular junction (site 4) and into the t-tubule (site 5) is detected by the dihydropyridine receptor (DHPR) that initiates the release of Ca^{2+} (Site 6) through the ryanodine receptors (RyR) on the SR membrane. This process requires proper ion gradients and maintained excitability determined, at least in part, by the Na^+/K^+ pump (site 3) and SR- Ca^{2+} pump activity (site 7). The binding of Ca^{2+} to the troponin-tropomyosin complex induces a series of conformational changes in the regulatory proteins (site 8) that allows the interaction of myosin and actin to initiate the power stroke (site 9). Contemporary cross-bridge theory depicted in (c) suggests that the chemo-mechanical transduction of the myosin-actin interaction is partitioned into ~6-strain-dependent structural transitions (steps 1–6) that make up the power stroke [32,53,72,73]. Briefly, starting in the rigor complex, ATP binds to the catalytic site on myosin and dissociates myosin from actin (step 1). The hydrolysis of ATP reprimers and cocks the myosin head which attaches to actin in a weakly bound state with the hydrolysis metabolites still bound to the catalytic site (step 2). In the conventional power stroke, the weakly bound cross-bridge transitions to the strongly bound state through unknown mechanisms (step 3). P_i is released from the catalytic site initiating the power stroke where myosin pivots at the light chain domain (step 4). The power stroke continues with an isomerization step where ADP is still bound to the catalytic site (step 5), followed by the release of ADP (step 6). Sites where elevated H^+ and P_i disrupt muscle contraction: A majority of the fatigue during high-intensity volitional exercise can be explained by the multifaceted and synergistic effects of increased levels of H^+ and P_i . Elevated H^+ disrupts contractile function by 1) reducing the sensitivity of the myofilaments to Ca^{2+} (site 8), 2) decreasing force of the cross-bridge by inhibiting the low- to high-force transition (step 3), and 3) slowing shortening velocity by inhibiting the ADP isomerization step (step 5) and/or the release of ADP (step 6). Elevated P_i disrupts contractile function by 1) decreasing the free $[Ca^{2+}]$ available for release from the SR (site 6), 2) reducing the sensitivity of the myofilaments to Ca^{2+} (site 8), and 3) decreasing force by inducing an unconventional power stroke where myosin dissociates from actin early in the high-force state before the release of P_i and ADP (step 4').

Figure 2



Associations between the accumulation of metabolites and the reductions in mechanical power during a fatiguing knee extension exercise.

Young (22.7 ± 1.2 yrs) and old participants (76.4 ± 6.0 yrs) performed a high-intensity fatiguing knee extension exercise consisting of 120 maximal velocity contractions (1 contraction per 2-s) lifting a load equivalent to 20% of the individual-specific maximal voluntary isometric contraction. The reduction in power during the 4-min dynamic fatiguing exercise was closely associated with the $[P_i]$ (a), pH (b), and concentration of diprotonated phosphate $[H_2PO_4^-]$ (c) measured over the final 64-s of the exercise. The greater fatigue in the old compared with young adults was accompanied by an ~30% greater increase in the $[H^+]$ (pH 6.61 versus 6.73) and an ~42% greater increase in the $[P_i]$ (32 versus 23 mM P_i). The combination of the greater decrease in pH and increase in $[P_i]$ resulted in an ~59% greater increase in the $[H_2PO_4^-]$ in the old compared with the young adults. The figure is reprinted with permission from Ref. [31**].

cellular and molecular preparations. In the remainder of the review, we focus primarily on recent discoveries of the effects of H^+ and P_i on contractile function from experiments using the chemically skinned fiber preparation and the *in vitro* motility and laser trap assays [10**,32,33**,34]. The advantage of these approaches is they permit precise control over the milieu surrounding the contractile proteins

to systematically study both the individual and combined effects of elevated metabolite levels on contractile function.

Effects of acidosis, H^+ , on contractile function

Although the role of H^+ in fatigue remains a topic of intense debate [35,36], the argument against acidosis as a putative mechanism of fatigue is based primarily on the modest effect H^+ has on isometric force under saturating Ca^{2+} conditions. Indeed, in saturating Ca^{2+} ($pCa = 4.5$ where $pCa = -\log_{10}[Ca^{2+}]$) a pH of 6.2 elicits a relatively small, albeit still significant, reduction in peak isometric force of 4–18% in skinned rat and rabbit fibers at 30°C [37,38*]. These findings are consistent with both the 10% decline in peak isometric force elicited by a pH of ~6.67 in living mouse muscle fibers at 32°C [39] and the 20% decline in force elicited by a pH of 6.5 in a mini-ensemble of myosin studied in a laser trap assay at 30°C [34]. The observation that elevated H^+ inhibits force production in both the laser trap assay with an unregulated thin filament [34] and in the skinned fiber preparation in saturating Ca^{2+} [37,38*] suggests that acidosis elicits decrements in force production, at least in part, by directly inhibiting the cross-bridge. While the mechanism remains unresolved, the acidosis induced decrements in isometric force may involve a reduction in the number of bound cross-bridges and/or the force generated per cross-bridge. The finding that the rate of force redevelopment (k_{tr}) following a slack re-extension maneuver of maximally Ca^{2+} -activated human fibers was slowed by elevated H^+ and P_i (pH 6.2 + 30 mM P_i) compared with a control condition (pH 7.0 + 4 mM P_i) suggests that acidosis may reduce the force per cross-bridge by inhibiting the low-force to high-force state of the cross-bridge cycle [10**]. This hypothesis is supported by the decreased high-force generating events and prolonged cross-bridge attachment times elicited by a pH of 6.5 in a mini-ensemble of myosin studied in a laser trap assay [34,40]. Thus, acidosis appears to inhibit peak isometric force in saturating Ca^{2+} primarily by reducing the force per cross-bridge via slowing the low-to high-force transition (step 3 Figure 1c) rather than reducing the number of bound cross-bridges.

In addition to the effect H^+ has on peak isometric force in saturating Ca^{2+} , intracellular acidosis contributes to fatigue by decreasing the sensitivity of the myofilament to Ca^{2+} . Because the relationship between the $[Ca^{2+}]$ and isometric force is sigmoidal, the acidosis-induced decrease in myofibrillar Ca^{2+} sensitivity manifests as a rightward shift in the force- pCa relationship and much greater reductions in isometric force when rat fibers were activated in pH 6.2 and submaximal compared with saturating Ca^{2+} [41]. The mechanisms for the decreased Ca^{2+} sensitivity are not fully elucidated; however, there is compelling evidence from skinned rabbit fibers that elevated H^+ reduces the affinity of the binding sites on troponin C (TnC) to Ca^{2+} [42,43] (site 8 Figure 1b). A subsequent study using the *in vitro* motility assay and a specific mutation in TnC that slows the release of Ca^{2+} from the binding sites revealed

that acidosis slows the rate that Ca^{2+} binds to TnC rather than accelerating the rate at which it's released [44]. In addition to the decreased Ca^{2+} affinity, there is also evidence that a H^+ -binding residue on troponin I (TnI) may contribute to the acidosis-induced decrease in Ca^{2+} sensitivity by altering the binding affinity of TnI to TnC [45]. Irrespective of the mechanism, the evidence that the free $[\text{Ca}^{2+}]$ decreases in the myoplasm during high-intensity contractions to subsaturating levels [46**] suggests that the H^+ -induced decrements in isometric force studied in saturating Ca^{2+} likely underestimate the depressive effects of acidosis on force production during fatigue *in vivo*.

Perhaps more important than the effect acidosis has on isometric force is the effect it has on shortening velocity and the ability to generate mechanical power (Figure 3). In saturating Ca^{2+} , a pH of 6.2 inhibits the maximal shortening velocity of skinned rat and rabbit fibers by 11–30% at 30°C, regardless of whether the velocity was measured with the slack test or extrapolation of the force–velocity curve [38*,47**]. These findings were corroborated by the observation that the actin filament velocity slowed markedly under acidic conditions in the *in vitro* motility assay at 20–30°C, which could be explained quantitatively by the acidosis-induced increase in the cross-bridge attachment times measured in the single molecule laser trap assay [33**,48]. The increased attachment times and slowed shortening velocity is thought to be due primarily to an inhibition of the ADP isomerization step of the cross-bridge cycle and/or the rate of ADP release [32,44] (steps 5 and 6 Figure 1c). The combination of the acidosis-induced decrease in both force and velocity resulted in an 18–34% reduction in peak power of rat fibers activated in pH 6.2 at 30°C [38*]. Thus, acidosis has a detrimental effect on contractile function by directly inhibiting force, velocity, and power of the cross-bridge and decreasing the sensitivity of the myofilament to Ca^{2+} .

Effects of inorganic phosphate, P_i , on contractile function

Similar to elevated H^+ , activating rat and rabbit fibers in saturating Ca^{2+} and 25–30 mM P_i reduced peak isometric force by 5–19% at 30°C [49,50]. However, the observation that the rate of force redevelopment following a slack re-extension maneuver of a maximally activated fiber (k_{tr}) is accelerated in the presence of elevated P_i [51,52] but slowed when H^+ and P_i are elevated together [10**] suggests that the mechanisms for the reduction in force differ for P_i compared with H^+ . While the mechanism is not fully elucidated, elevated P_i is thought to inhibit force, at least in part, by inducing an unconventional power stroke where myosin dissociates from actin early in the high-force state of the cross-bridge cycle before the release of P_i and ADP [32,53,54*] (step 4' Figure 1c). This mechanism is consistent with the increased ATP cost of contraction observed in skinned fibers that occurs from

the P_i -induced decrease in isometric force but a maintained myofibrillar ATP hydrolysis rate [54*,55]. Further support for this mechanism comes from the observation that 30 mM P_i elicited a 65% decrease in the cross-bridge attachment times of a mini-ensemble of myosin studied in a laser trap assay at 30°C [56*]. Thus, elevated P_i appears to inhibit peak isometric force in saturating Ca^{2+} primarily by accelerating the detachment of myosin from actin, which reduces the number of bound cross-bridges.

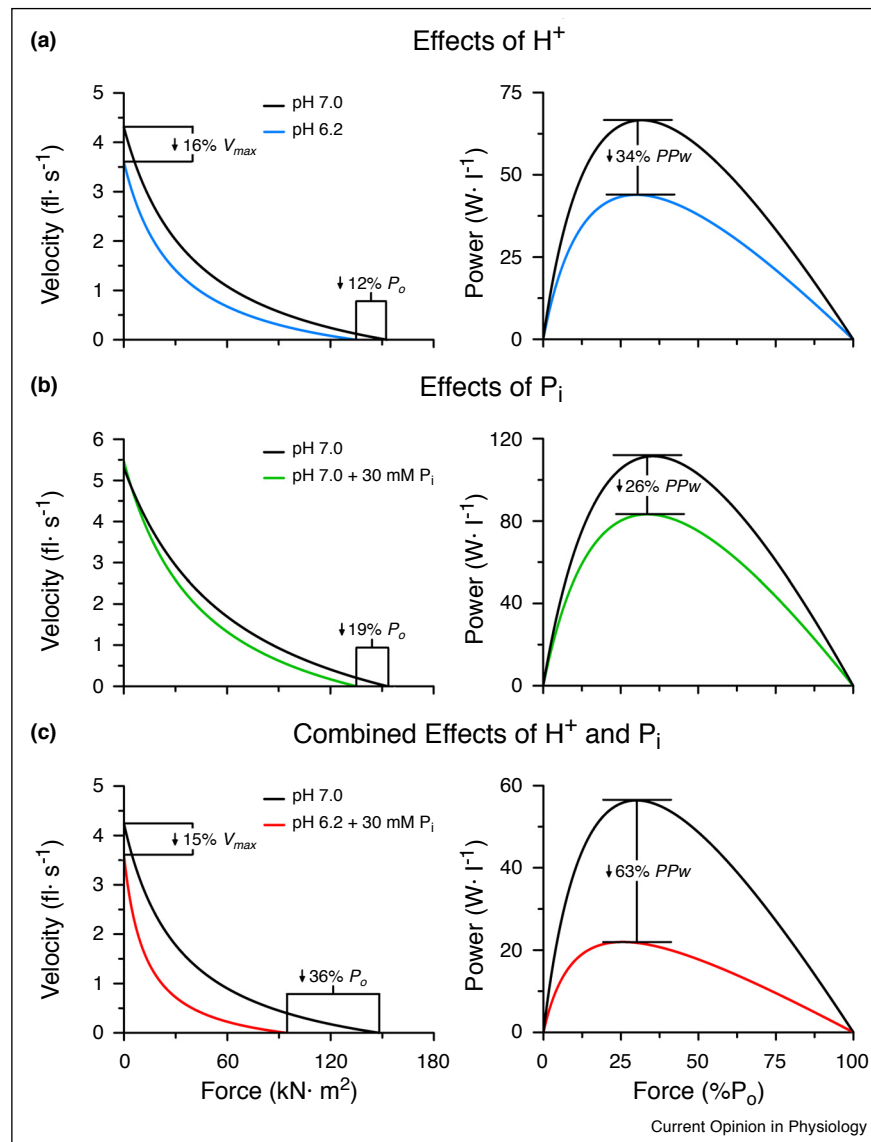
In addition to the effect P_i has on isometric force in saturating Ca^{2+} , elevated P_i contributes to fatigue by decreasing the sensitivity of the myofilament to Ca^{2+} [57] and inhibiting Ca^{2+} release from the sarcoplasmic reticulum (SR) [13,58,59]. The mechanism for the P_i -induced decrease in myofibrillar Ca^{2+} sensitivity is unknown and needs further investigation; however, it is clear that, unlike H^+ , elevated P_i does not alter the affinity of the binding sites on TnC to Ca^{2+} [42]. In contrast to the lack of understanding of how elevated P_i decreases myofibrillar Ca^{2+} sensitivity, there is more evidence describing the mechanism by which elevated P_i inhibits Ca^{2+} release from the SR [13,58,60]. Briefly, P_i is thought to reduce the amount of free $[\text{Ca}^{2+}]$ available for release from the SR by diffusing into the SR through a P_i -permeable channel [61] and binding to Ca^{2+} to form a precipitate [58–60,62] (site 6 Figure 1b). Similar to H^+ , the reduced free $[\text{Ca}^{2+}]$ available in the myoplasm to subsaturating levels coupled with the decreased myofibrillar Ca^{2+} sensitivity suggests that the P_i -induced decrements in isometric force studied in saturating Ca^{2+} likely underestimate the depressive effects of P_i on force production during fatigue *in vivo*.

Unlike elevated H^+ , P_i does not alter the shortening velocity of rat, rabbit or human fibers activated at 30°C [10**,47**,50]. However, athletic prowess and the ability to perform daily activities are determined more by the muscle's ability to generate mechanical power, and 30 mM P_i elicited an 18–26% reduction in peak power in rat fibers activated at 30°C [50] (Figure 3). Thus, elevated P_i has a detrimental effect on contractile function by directly inhibiting force and power of the cross-bridge and by decreasing both the free $[\text{Ca}^{2+}]$ available for release from the SR and the sensitivity of the myofilament to Ca^{2+} .

Synergistic effects of H^+ and P_i on contractile function

Although studies on the individual effects of H^+ and P_i are important, studying their effects when elevated together is more pertinent to understanding how these metabolites contribute to the decrements in contractile function that occur during high-intensity fatiguing exercise *in vivo* [17*,19,20,31**]. Given the evidence that elevated H^+ and P_i contribute to fatigue by different mechanisms, it

Figure 3



Individual and combined effects of elevated H^+ and P_i on force, velocity, and power of rat slow MHC I fibers at 30°C.

Force-velocity and force-power curves obtained from rat MHC I fibers in the control pH 7.0 condition compared with pH 6.2 (a), 30 mM P_i (b), and combined pH 6.2 and 30 mM P_i (c) at 30°C. Shortening velocity (fiber lengths per second) and power ($W \cdot l^{-1}$) are plotted as a function of the force expressed relative to the fiber cross-sectional area ($kN \cdot m^2$) and as a percentage of the peak isometric force (% P_o), respectively. Error bars around the mean curves have been omitted for clarity. Findings from fast fibers are qualitatively similar to those presented here. The figure is reprinted with permission from Ref. [32] and the data obtained from Refs. [38*,50,63*].

is perhaps not surprising that when studied in combination their depressive effects on contractile function are additive. For example, a combined pH 6.2 and 30 mM P_i condition decreased peak isometric force of rat fibers by 36–46% in saturating Ca^{2+} at 30°C [63*] and caused a considerably greater decrease in the sensitivity of the myofilament to Ca^{2+} than either metabolite alone [41,57]. Interestingly, the combined condition had a synergistic effect where the decrements were greater than would be predicted from the sum of the individual metabolite

effects, particularly for peak power of rat and rabbit fast fibers [38*,47**,50,63*]. The combined pH 6.2 and 30 mM P_i condition decreased peak power by 55–63% in rabbit and rat fibers at 30°C (Figure 3), which was exacerbated to a 70% decline in power in rabbit fast fibers when the myosin regulatory light chain was phosphorylated [47**,63*]. Extrapolating these findings to the whole-muscle suggests that ~55–70% of the reduction in the ability to generate power can be attributed to the synergistic effects of elevated H^+ and P_i directly inhibiting the

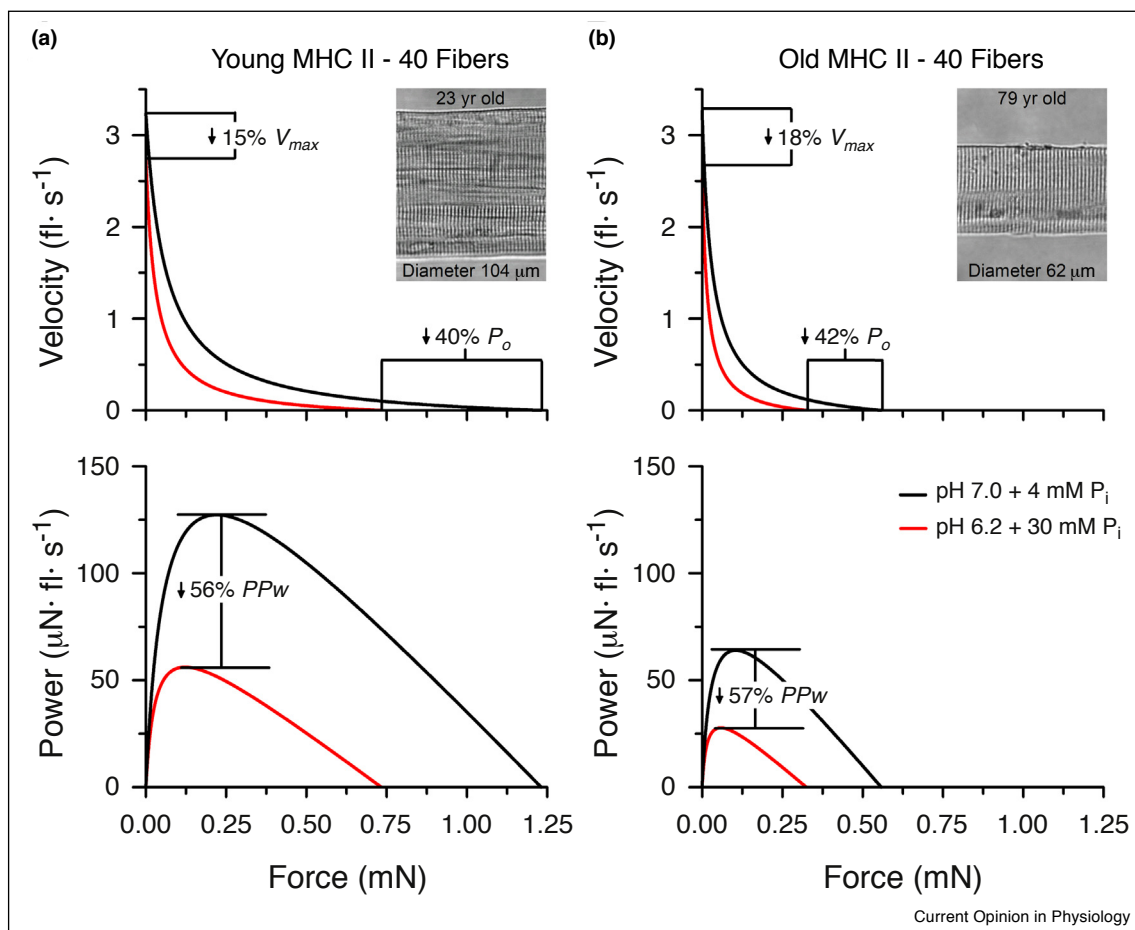
cross-bridge, which is likely exacerbated by the effects these metabolites also have on myofibrillar Ca^{2+} sensitivity and the release of Ca^{2+} from the SR. The mechanisms to explain the synergistic effects are unknown [34,40], but may include alterations in the binding affinities of the myosin-actin interaction at different steps in the cross-bridge cycle [64].

Translating the bioenergetics perspective to understanding fatigue in old adults

The ability of old adults to generate power is severely compromised by the combination of the atrophy of muscle fibers expressing the fast myosin heavy chain isoforms [10[•],65,66] and the increased fatigue that occurs when old adults perform moderate- to high-velocity

contractions [11,67–70]. The mechanisms for the age-related increase in fatigue are unresolved; however, studies employing non-invasive stimulation procedures to the intact neuromuscular system have localized the primary site to within the muscle rather than the nervous system [10[•],11,68,69,71]. Translating the understanding of the bioenergetic basis of fatigue to this problem, we tested whether age-related changes of the muscle resulted in either 1) an increased sensitivity of the cross-bridge to a given concentration of metabolite accumulation or 2) an increased production of metabolites due to a greater reliance on anaerobic metabolism. To test these hypotheses, we exposed fibers from the vastus lateralis of young and old men to a condition mimicking quiescent skeletal muscle (pH 7.0 + 4 mM P_i) and the

Figure 4



Force-velocity and force-power curves of fast MHC II fibers from young and old men at 15°C.

Absolute peak power (PPW) and peak isometric force (P_o) of the fast MHC II fibers from young adults (a) were ~twofold greater than in fibers from the old adults (b). The age differences in absolute PPW and P_o were explained entirely by differences in the size of the fibers between young and old adults. The insets are digital images taken of young and old adult fast fibers at 800 $\times$ magnification that are representative of the average size of the fast MHC II fibers studied in the young and old adults. The combined pH 6.2 and 30 mM P_i condition caused significant decreases in V_{max} , P_o , and PPW compared to the condition mimicking quiescent human skeletal muscle (pH 7.0 + 4 mM P_i) in fibers from young and old adults; however, the relative reductions did not differ with age. The variances around the mean curves were omitted for clarity. The figure was modified with permission from Ref. [10[•]].

combined pH 6.2 and 30 mM P_i condition (Figure 4). While the data confirmed that these metabolites act synergistically to impair cross-bridge function, the decrements in force, velocity, and power from the combined pH 6.2 and 30 mM P_i condition did not differ in the fibers isolated from young compared with old men [10^{••}]. In contrast, when we had young and old adults perform a dynamic fatiguing knee extension exercise while simultaneously measuring the intracellular metabolite accumulation with ^{31}P -MRS, the greater fatigue in the old compared with young adults was accompanied by an $\sim 30\%$ greater increase in the $[H^+]$ (pH 6.61 versus 6.73) and an $\sim 42\%$ greater increase in the $[P_i]$ (32 versus 23 mM P_i) [31^{••}]. Importantly, the reductions in power during the fatiguing exercise were closely associated with the intracellular metabolite accumulation (Figure 2) suggesting that the increased fatigue in old adults during dynamic exercise has a bioenergetic basis explained by an increased accumulation of metabolites within the muscle [31^{••}].

Concluding remarks

Integrating findings from *in vivo* and isolated cellular and molecular studies has provided considerable advancements in our understanding of how a reliance on anaerobic metabolism disrupts contractile function during high-intensity exercise. While many of the mechanisms are not fully resolved, the data suggest that fatigue during high-intensity exercise is determined, in large part, by the rate and extent of intracellular metabolite accumulation. We conclude that a majority of the fatigue in healthy young and old adults performing high-intensity volitional exercise can be explained by the multifaceted and synergistic effects of elevated H^+ and P_i acting to impair contractile function within the muscle. Translating this understanding of the bioenergetic basis of fatigue to clinical populations may help guide studies aimed at identifying the mechanisms of fatigue and designing targeted therapies to offset the detrimental effects of fatigue in these alternative populations. Future studies are needed to identify how elevated H^+ and P_i act synergistically to impair cross-bridge function and decrease myofibrillar Ca^{2+} sensitivity so that we may ultimately develop treatments to attenuate the effects of these metabolites on contractile function.

Conflict of interest statement

Nothing declared.

Author contributions

C.W.S. prepared the figures and drafted the manuscript; C.W.S. and R.H.F. edited and revised the manuscript. Both authors approved the final version of the manuscript and agree to be accountable for all aspects of the work.

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Papers of particular interest, published within the period of review, have been highlighted as:

- of special interest
- of outstanding interest

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This is the first study to examine the combined effects of elevated H^+ and P_i on contractile function of fibers isolated from human skeletal muscle. The data confirmed previous non-human studies that the metabolites act synergistically to impair cross-bridge function, but that the effects were similar in fibers isolated from young and old men. The findings highlight the importance of the metabolites in human muscle fatigue, while also suggesting that the age-related increase in fatigue during dynamic

exercise cannot be explained by an increased sensitivity of the cross-bridge to H^+ and P_i .

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